**Aaron Bernstein**
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## EDUCATION

**Penn State University**, State College, PA *Anticipated graduation:* May 2027

- **Bachelor of Science in Computer Science, Minor in AI Engineering | GPA: 3.46**
- ***Relevant Coursework*:** Data Structures & Algorithms, Machine Learning & AI, Systems Programming, Object-Oriented Programming with Web Applications, Programming Language Concepts, Computer Organization & Design, Discrete Mathematics, Probability & Statistics, Exploratory Data Mining and Theory of Computation
- ***Honors:*** *Dean's List (Fall 2024, Spring 2025, Fall 2025)*

## SKILLS

**Languages:** C, Java, Python, SQL
**Technologies:** TypeScript, React, Node.js, Cloudflare Workers and D1, Drizzle ORM, tRPC, PyTorch, pandas, scikit-learn
**Tools:** Git/GitHub, Claude Code, VS Code, Linux/Unix CLI

## EXPERIENCE

**Snapback Sports**, New York City, NY June 2026 - Present

*Software Engineer, Part-Time (August 2026 - Present)*

- Maintain the Snapback games hub in production, handling upkeep and fixes across the platform's seven games
- Serve as the primary tester for new products ahead of release, writing detailed feedback reports
- Add value in the eyes of sponsors/partners by creating websites specific to Snapback's partnerships

*AI Intern (June 2026 - July 2026)*

- Shipped a four-game daily sports trivia platform as sole engineer on React Router v7, Cloudflare Workers, and Drizzle ORM over D1, then integrated it into a seven-game hub unifying all intern-built games under one login
- Co-built and launched a college football gameday review platform (TanStack Start, Cloudflare Workers, D1, R2) with a three-person team, reaching 100+ users within a week of launch
- Managed and created videos and social media posts, generating 24 million impressions across all platforms promoting the internship program and products we shipped

**Alphawave Semi**, San Jose, CA July 2025 - August 2025

*Semiconductor chip testing intern*

- Performed environmental stress testing on semiconductor chips, validating chip performance under different conditions to verify and track failure points
- Analyzed and documented error logs within Excel, categorizing errors to assist senior engineers in debugging and analysis

## PROJECTS

**BlueLineTradingGroup.com**

- Designed and shipped a five-page site for a plumbing supply distributor, replacing an aging WordPress build; live in production as the company's website
- Hardened the contact API with server-side validation, a strict field allowlist, IP-based rate limiting, and a honeypot, then ran a file-by-file security audit that surfaced an email header injection vector (CRLF sequences reaching the subject line) and a spoofable rate-limit bypass, fixing both
- Built the inquiry pipeline end to end: a Next.js API route that validates and sanitizes submissions before relaying them over authenticated SMTP through a dedicated sending account

**March Madness model (github.com/AaronZacharyBernstein/MarchMadnessModel)**

- Assembled a 17-season dataset (2008-2025, 544 tournament teams) of advanced college basketball metrics from public analytics sources including BartTorvik and KenPom
- Built leave-one-season-out cross-validation against a seed-only baseline, which revealed the initial 22-feature random forest ranked the eventual champion first in only 4 of 17 held-out seasons
- Cut to six features, normalized each stat within its own season to remove era effects, and replaced a forest with regularized logistic regression, raising top-1 accuracy to 8/17 and top-3 to 14/17 against baselines of 5 and 12